



VERDICT The five challenge claims are **faithfully reported in arXiv v1**, but Claims 1–3 cannot be rerun because the trained 7B checkpoint, evaluator, RL code, and trajectories remain unreleased. Claims 4–5 are independently executable at mechanism level.

3 / 5
benchmark claims
source-only

5,000
random subsets
in GPU proxy

2 / 5
mechanisms
executable

v1 ≠ v2
material equation
and table drift

1 Bottom line ★ HEADLINE

This is a **partial reproduction**: exact paper/source verification, an executable diversity proxy, and reward-equation tests—not a rerun of ML-Agent’s benchmark superiority.

7B claimed agent size 9 reported train tasks 10 core held-out tasks

Read claims by evidence tier: reported, source-verified, reproduced, or proxy-only.

2 Release gate

Asset	Public?	Effect
Trained checkpoint	No	No agent run
Evaluator	No	No Table 1 rerun
Step-wise RL	No	No training replay
Trajectories	No	No data audit
SFT / eval scripts	Stubs	“To be released”

Official commit 15932e7, audited 16 July 2026. Missing artifacts block evidence; they do not falsify the reported scores.

3 Evidence policy

- **Source-verified**: statement located in v1 or v2.
- **Reproduced**: independent executable check.
- **Proxy**: disclosed substitute data or embedding model.

3a Primary sources

- **Paper**: arXiv 2505.23723 v1 and v2.
- **Review**: OpenReview kcPPWaoegr.
- **Code**: MASWorks/ML-Agent at commit 15932e7.

Poster hotspots open the matching Trackio evidence page; archived PDFs and Job outputs are bundled.

4 Claim 1 · Model scale

Verified in v1 Table 1: the 7B Qwen2.5-based ML-Agent is reported above the listed GPT-4o, GPT-4o-mini, and 671B DeepSeek-R1 agent baselines across the evaluation suite. Revision warning: v2 replaces the GPT-4o comparators with GPT-5, Gemini, and Qwen3 agents; the challenge wording is v1-specific.

5 Claim 2 · Generalization

Split verified: 4 MLAGentBench plus 5 MLE-bench tasks for training; 10 held-out MLE-bench tasks cover image, text, and tabular machine-learning problems.

- The 9-task count is explicit in both revisions.
- Task tables substantiate all three modalities.
- The release lacks both weights and trajectories for an independent held-out evaluation.

Status: split verified; generalization result blocked.

6 Claim 3 · CIFAR-10

68.88% v1 average accuracy 81.45% v1 best of eight 33.80 v2 mean gain

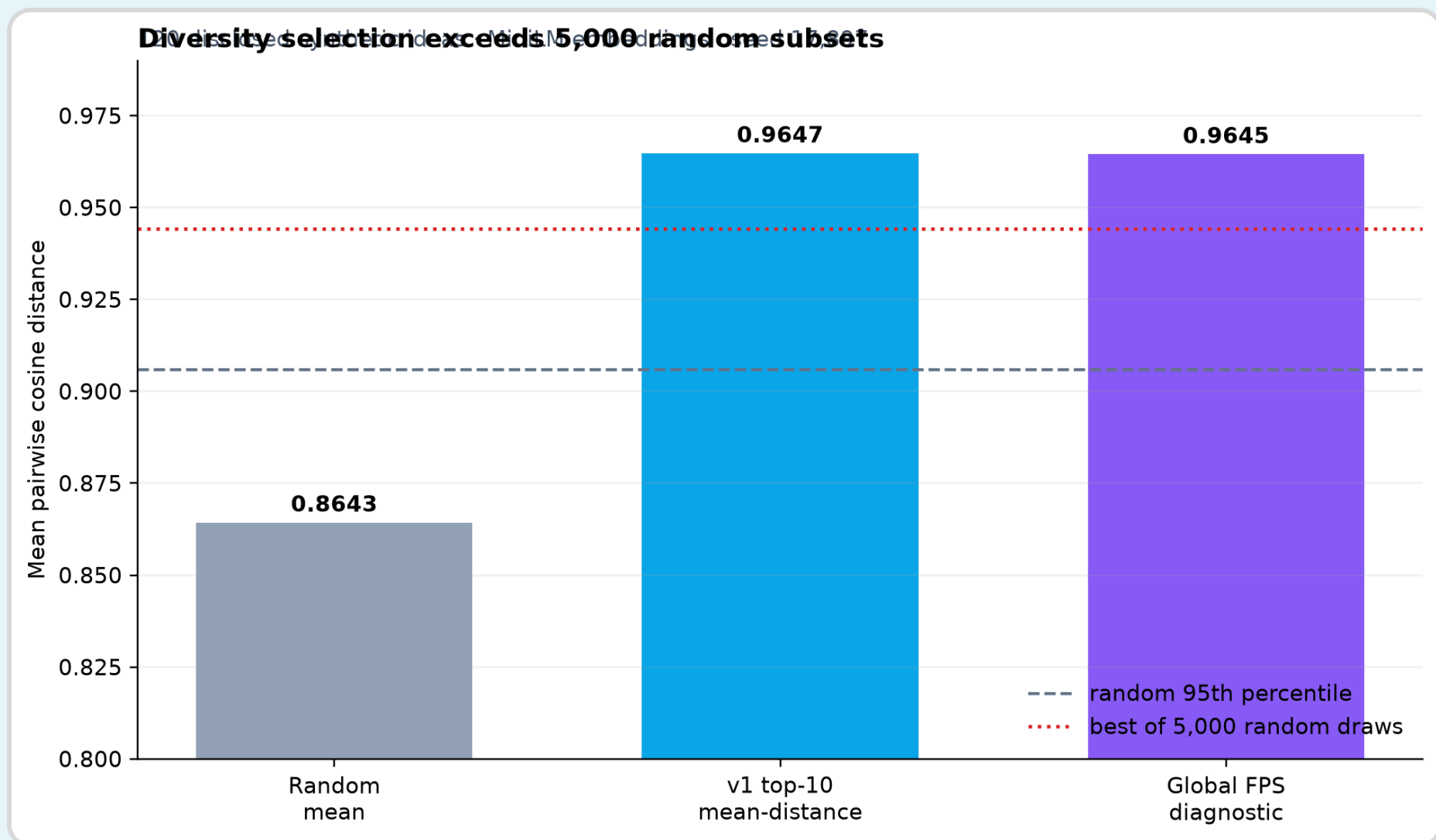
The requested absolute accuracies are exact v1 Table 1 values. V2 instead reports 33.80 ± 11.27 percentage performance gain.

6a Revision map

Item	v1	v2
Comparators	GPT-4o	GPT-5 / Gemini
CIFAR	Accuracy	Gain
Reward	Sigmoid	Linear

The challenge uses the archived v1 framing; later revisions change the comparison, so claims must be version-pinned.

7 Claim 4 · Diversity selection ★ REPRODUCED



Independent T4 proxy. 120 disclosed synthetic ideas, MiniLM embeddings, seed 17,897. The v1 top-10 selector exceeds every one of 5,000 random subsets.

0.9647 selected versus 0.8643 random mean and a 0.9443 maximum across the sampled baseline.

8 What Claim 4 establishes

- **v1 source**: generate at least 100 ideas, keep 10 with greatest mean embedding distance.
- **Proxy result**: the reconstructed selector strongly enriches pairwise diversity on disclosed data.
- **Not established**: diversity of the unreleased author pool or downstream RL performance.

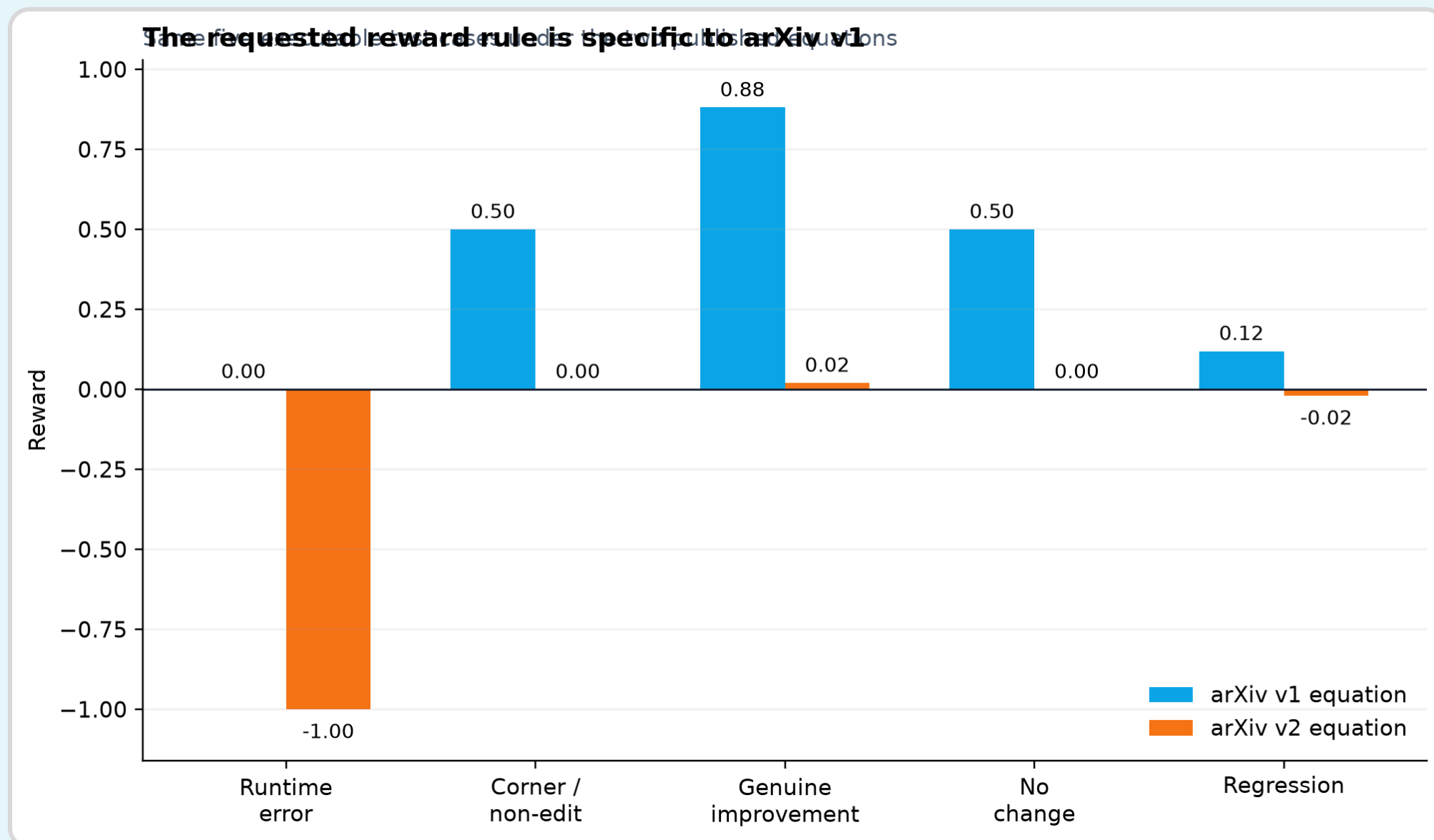
Global FPS improves the minimum from 0.2034 to 0.8772; v2’s per-axis FPS remains a different procedure.

9 Compute audit

- Completed Job: T4-small, 48 s running, 64 s wall time.
- All attempts: 138 GPU-running seconds, about USD 0.0153; conservative bound below USD 0.02.

Job: Hugging Face 6a5830b6b1669a49bf0767d3.

10 Claim 5 · Reward ★ REPRODUCED



Executable branch tests. The requested 0 / 0.5 / sigmoid rule is the archived v1 equation; v2 uses −1 / 0 / linear normalized change. A +0.01 improvement yields 0.8808 in the v1 scaling example, compared with 0.02 under v2.

11 Reward rule changed

Trajectory	v1	v2
Invalid / error	0	−1
Corner / non-edit	0.5	0
Improvement	Sigmoid	Linear
Regression	0.1192	−0.02

The challenge statement is correct for v1 and incorrect if presented as the current v2 formula.

12 What would unlock a full rerun

- Publish the trained Qwen2.5-7B checkpoint.
- Release evaluator, task environments, and exact prompts.
- Release SFT/RL code plus training trajectories and seeds.

Take-home: mechanisms are coherent and testable; headline superiority remains paper-reported evidence.